

Circuit Simulation Lab:25

Frequency Dividers

The purpose of this experiment is to introduce the design of Divide by 2 Circuit

D-type Flip-Flop is as a binary divider, for Frequency Division or as a “divide-by-2” counter. Here the inverted output terminal Q (NOT-Q) is connected directly back to the Data input terminal D giving the device “feedback” as shown below.

CODE:

```
module divide_by_2(
input clk,
input rst,
output reg q
);

always @(posedge clk)
begin

if(rst)
    q <= 1'b0;

else
    q <= ~q;

end

Endmodule
TESTBENCH
module tb_divide_by_2;

reg clk,rst;
wire q;

divide_by_2 dut(
.clk(clk),
.rst(rst),
.q(q)
);

always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("divide_by_2.vcd");  
$dumpvars(0,tb_divide_by_2);
```

```
clk=0;  
rst=1;
```

```
#20 rst=0;
```

```
#200;
```

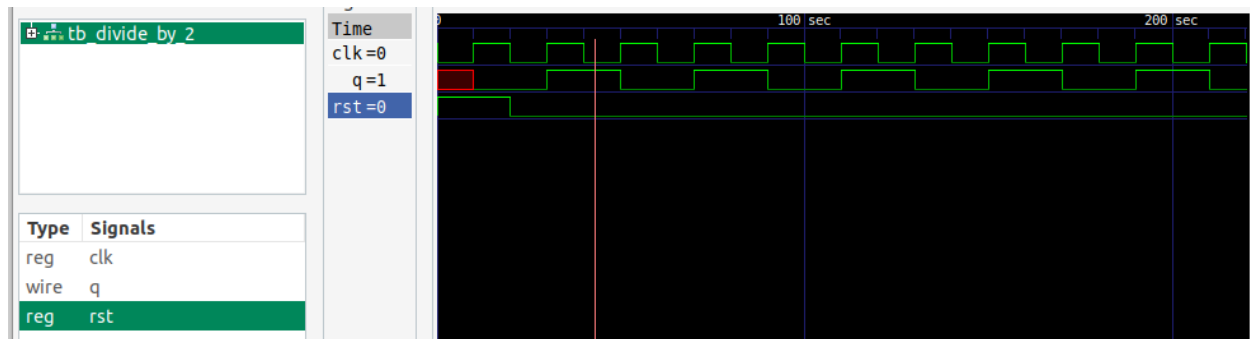
```
$finish;
```

```
end
```

```
initial
```

```
$monitor("time=%0t clk=%b q=%b",  
        $time,clk,q);
```

```
endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_25$ iverilog -o out
divide_by_2.v tb_divide_by_2.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_25$ vvp out
VCD info: dumpfile divide_by_2.vcd opened for output.
time=0 clk=0 q=x
time=10 clk=1 q=0
time=20 clk=0 q=0
time=30 clk=1 q=1
time=40 clk=0 q=1
time=50 clk=1 q=0
time=60 clk=0 q=0
time=70 clk=1 q=1
time=80 clk=0 q=1
time=90 clk=1 q=0
time=100 clk=0 q=0
time=110 clk=1 q=1
time=120 clk=0 q=1
time=130 clk=1 q=0
time=140 clk=0 q=0
```